



**Basic Tutorial**

# Home Module

Home Module

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Home Module

|    |    |    |    |    |    |    |
|----|----|----|----|----|----|----|
|    | S1 | S2 | S3 | S4 | S5 | S6 |
| P1 | 5  | 16 | 5  | 0  | 0  | 3  |
| P2 | 21 | 45 | 58 | 14 | 10 | 15 |

Peak Count Data

C

Data Filtering and Grouping Module

Group1

S1

S2

S3

Group2

S4

S5

S6

D

Differential Peak Analysis Module

|    | log2FC | pvalue | padj |
|----|--------|--------|------|
| P1 | -2.93  | 0.04   | 0.39 |
| P2 | -1.35  | 0.02   | 0.26 |

Differential Peak Table

B

Peaks Visualization Module

Coverage Plot

Profile Plot

E

Differential Peak Annotation Module

|    | anno   | TSS    | geneID |
|----|--------|--------|--------|
| P1 | Exon   | 779943 | 114798 |
| P2 | Intron | 283818 | 2567   |

Peak Annotation Table

Upset Plot

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Differential Peak Visualization Module

Group1 vs Group2

Volcano Plot

Heatmap

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Functional Enrichment Module

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Motif Enrichment Module

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Import Data

Data Source:

Example dataset

Example dataset

Upload data

dexamethasone treatment, obtained from ENCODE (ENCSR783SNV and ENCSR543ZVZ). Each condition contains three biological replicates, and the peaks have been pre-processed into a merged count matrix ready for analysis. You can load this dataset to explore the features and workflow of ChromTag without uploading your own files.

START

Instruction

The Home Module serves as the entry point of ChromTag, providing an integrated interface for initializing data input and establishing the foundation for subsequent analyses. Users may choose to begin with the system-provided example dataset or upload custom peak count matrices that have been pre-merged across samples. This module also manages species selection and displays an immediate preview of the imported dataset, enabling users to verify data structure and completeness before progressing to downstream analytical modules. Together, these components ensure that the analysis pipeline begins with properly formatted and biologically consistent input data, thereby supporting accurate and reliable interpretation in later stages.

Users can choose between “**Example Dataset**” and “**Upload Data**”.

If the **Example Dataset** is selected, the application automatically loads a **pre-processed merged peak count matrix** containing H3K27ac ChIP-seq profiles from A549 cells collected at 0 h and 4 h after dexamethasone treatment. The dataset was obtained from ENCODE, including ENCSR783SNV and ENCSR543ZVZ, with **three biological replicates** for each condition.

# Home Module

 Import Data

Data Source:

Upload data

Select Species:

Human

Upload CSV File

Browse...

No file selected

If you choose to upload custom data, you must also select the appropriate species.

| Chromosome | Start  | End    | A549_0h_rep1 | A549_0h_rep2 | A549_0h_rep3 | A549_4h_rep1 | A549_4h_rep2 | A549_4h_rep3 |
|------------|--------|--------|--------------|--------------|--------------|--------------|--------------|--------------|
| chr1       | 629727 | 630151 | 31           | 12           | 5            | 19           | 18           | 23           |
| chr1       | 633792 | 634272 | 44           | 24           | 10           | 17           | 48           | 25           |
| chr1       | 778021 | 778694 | 46           | 54           | 59           | 69           | 26           | 64           |
| chr1       | 778877 | 779438 | 69           | 53           | 57           | 66           | 37           | 70           |
| chr1       | 779666 | 779906 | 20           | 18           | 22           | 23           | 12           | 29           |
| chr1       | 826707 | 827761 | 97           | 90           | 94           | 88           | 47           | 100          |
| chr1       | 870010 | 870297 | 9            | 17           | 4            | 12           | 6            | 12           |
| chr1       | 904028 | 905405 | 205          | 231          | 182          | 178          | 107          | 185          |
| chr1       | 906957 | 907195 | 28           | 31           | 38           | 18           | 13           | 24           |

Uploaded data should be **pre-merged across samples** and must follow a specific format. The file should be a **CSV** with a header row.

The first column must contain the chromosome name (e.g., “chr1”), the second and third columns should specify the **start** and **end** positions, and all subsequent columns should represent the **peak count values** for each sample.

Each experimental group must include **at least two biological replicates** to ensure proper statistical analysis.

# Home Module

Import Data

Data Source:

Example dataset

Example dataset

Upload data

dexamethasone treatment, obtained from ENCODE (ENCSR783SNV and ENCSR543ZVZ). Each condition contains three biological replicates , and the peaks have been pre-processed into a merged count matrix ready for analysis. You can load this dataset to explore the features and workflow of ChromTag without uploading your own files.

▶ START

After choosing the data source, users can click the “**Start**” button to load the dataset and proceed to the next module.

Data Preview

Show

10

entries

Search:

|    | Chromosome | Start  | End    | A549_0h_rep1 | A549_0h_rep2 | A549_0h_rep3 | A549_4h_rep1 | A549_4h_rep2 | A549_4h_rep3 |
|----|------------|--------|--------|--------------|--------------|--------------|--------------|--------------|--------------|
| 1  | chr1       | 629727 | 630151 | 31           | 12           | 5            | 19           | 18           | 23           |
| 2  | chr1       | 633792 | 634272 | 44           | 24           | 10           | 17           | 48           | 25           |
| 3  | chr1       | 778021 | 778694 | 46           | 54           | 59           | 69           | 26           | 64           |
| 4  | chr1       | 778877 | 779438 | 69           | 53           | 57           | 66           | 37           | 70           |
| 5  | chr1       | 779666 | 779906 | 20           | 18           | 22           | 23           | 12           | 29           |
| 6  | chr1       | 826707 | 827761 | 97           | 90           | 94           | 88           | 47           | 100          |
| 7  | chr1       | 870010 | 870297 | 9            | 17           | 4            | 12           | 6            | 12           |
| 8  | chr1       | 904028 | 905405 | 205          | 231          | 182          | 178          | 107          | 185          |
| 9  | chr1       | 906957 | 907195 | 28           | 31           | 38           | 18           | 13           | 24           |
| 10 | chr1       | 910946 | 911195 | 18           | 17           | 15           | 13           | 6            | 21           |

Showing 1 to 10 of 76,176 entries

Previous

1

2

3

4

5

...

7,618

Next

Once the data is successfully loaded, the system will display the **peak count matrix** in a preview table for inspection.

# Peaks Visualization Module

This step is optional and can be skipped.

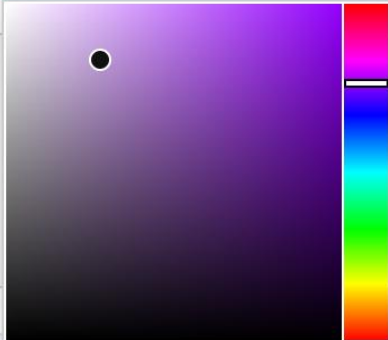
 Assign Sample Colors

Sample:

H3K4me3\_rep2

Choose a Color:

#BE9CD6



✓ Submit

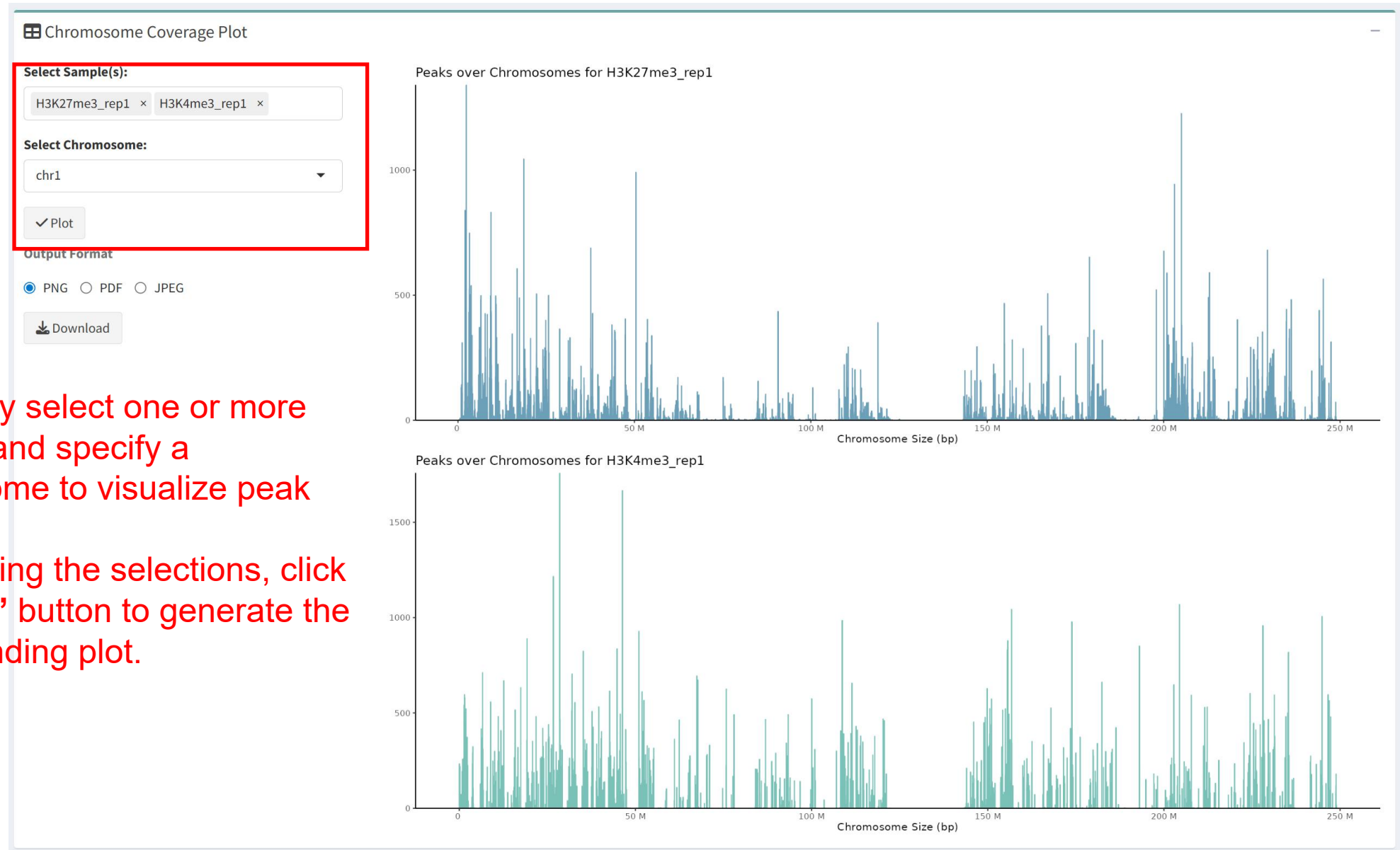
| Sample        | Color   |
|---------------|---------|
| H3K27me3_rep1 | #6C9FB8 |
| H3K27me3_rep2 | #A587C9 |
| H3K4me3_rep1  | #7AC2B8 |
| H3K4me3_rep2  | #C5D192 |

Users can select a sample and assign a preferred color using the color picker. Once submitted, the chosen color will be applied consistently across the **Chromosome Coverage Plot** and the **TSS Profile Plot**, enhancing visual distinction among samples.

# Peaks Visualization Module

This step is optional and can be skipped.

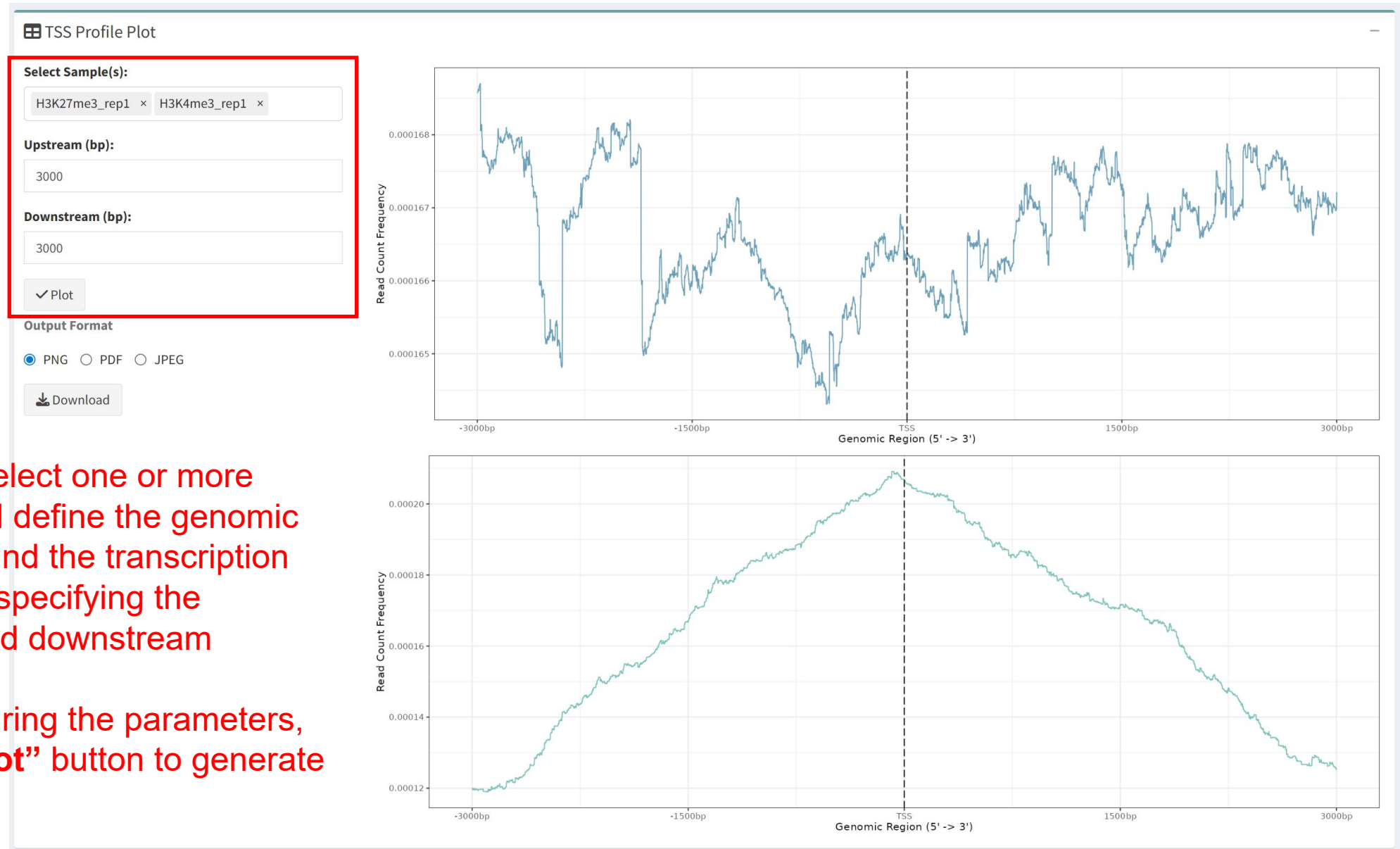
Users may select one or more samples and specify a chromosome to visualize peak coverage. After making the selections, click the “**Plot**” button to generate the corresponding plot.





# Peaks Visualization Module

This step is optional and can be skipped.



Users can select one or more samples and define the genomic window around the transcription start site by specifying the upstream and downstream distances. After configuring the parameters, click the “**Plot**” button to generate the plot.

# Data Filtering and Grouping Module

Before Filter

After Filter

Count Threshold: ?

5

✓ Submit

Show

5

entries

Search:

| hromosome | Start  | End    | H3K27me3_rep1 | H3K27me3_rep2 | H3K4me3_rep1 | H3K4me3_rep2 |
|-----------|--------|--------|---------------|---------------|--------------|--------------|
| 1r1       | 27805  | 30656  | 0             | 0             | 233          | 186          |
| 1r1       | 135727 | 140083 | 0             | 2             | 166          | 89           |
| 1r1       | 198335 | 200968 | 0             | 0             | 219          | 169          |
| 1r1       | 391014 | 392146 | 1             | 3             | 0            | 1            |
| 1r1       | 392617 | 393317 | 2             | 1             | 0            | 0            |

Showing 1 to 5 of 220,230 entries

Previous

1

2

3

4

5

...

44,046

Next

Users may specify a threshold value to filter the data.

During filtering, the system sums the read counts for each genomic region across samples and retains only those entries whose total count exceeds the defined threshold.

> 5

< 5



# Data Filtering and Grouping Module

## ↔ Sample Grouping

### Select Sample(s):

A549\_4h\_rep1 × A549\_4h\_rep2 × A549\_4h\_rep3 ×

Select the samples that should be grouped together.

### Select Group Label:

Experimental group

Select “**experimental group**” or “**control group**” as the group name.

Add Group

Clear Groups

Use the “**Add Group**” button to create the group.

Use the “**Clear Groups**” button to remove all existing groups.

### Current Groups:

\$Control  
[1] "A549\_0h\_rep1" "A549\_0h\_rep2" "A549\_0h\_rep3"

All defined groups are listed under “**Current Groups.**”

✓ Submit

After all groups have been defined, click the “**Submit**” button to save the grouping information.

# Differential Peak Analysis Module

⬅ Differential Analysis Parameters !

Direction of Change: ?  
Both Directions ▼

Apply Independent Filtering for Low-Count Peaks  
? ☒ Yes ☐ No

Independent filtering target adjusted p-value: ?  
0.01

P-value adjustment method:  
Benjamini–Hochberg (BH) ▼

▶ Run Differential Analysis

## Direction of Change

This setting lets users choose whether to detect peaks that are increased or decreased between groups, allowing the analysis to focus on a specific direction of change.

## Independent Filtering

Independent filtering is a DESeq2 step that ignores low-signal peaks during multiple-testing correction to improve detection power. Filtered peaks remain in the results, but their adjusted p-values may be NA.

## target adjusted p-value

DESeq2 uses this value to choose a mean-count filtering threshold that helps maximize the number of significant peaks after multiple-testing correction. It does not directly define the final differential peaks; final peak selection is still based on the adjusted p-value and log2FC cutoffs specified later.

## p-value Adjustment Method

This option allows users to select a method for multiple-testing correction.

- The Benjamini–Hochberg (BH) method controls the false discovery rate (FDR) and is well suited for analyses with many tests.
- The Holm method controls the family-wise error rate (FWER) and is more conservative, reducing false positives but potentially missing true signals.

# Differential Peak Annotation Module

Input Annotation Parameters

Upstream (bp):

3000

Downstream (bp):

3000

▶ Run Peak Annotation

To annotate the peaks, first specify the upstream and downstream distances to define the genomic region of interest, and then click the “Run Peak Annotation” button to execute the annotation.

# Differential Peak Visualization Module

 Volcano Plot

 MA Plot

 PCA Plot

 Heatmap

-log10(padj) Cutoff

1.3

Log2FC Threshold (Positive)

1

Log2FC Threshold (Negative)

1

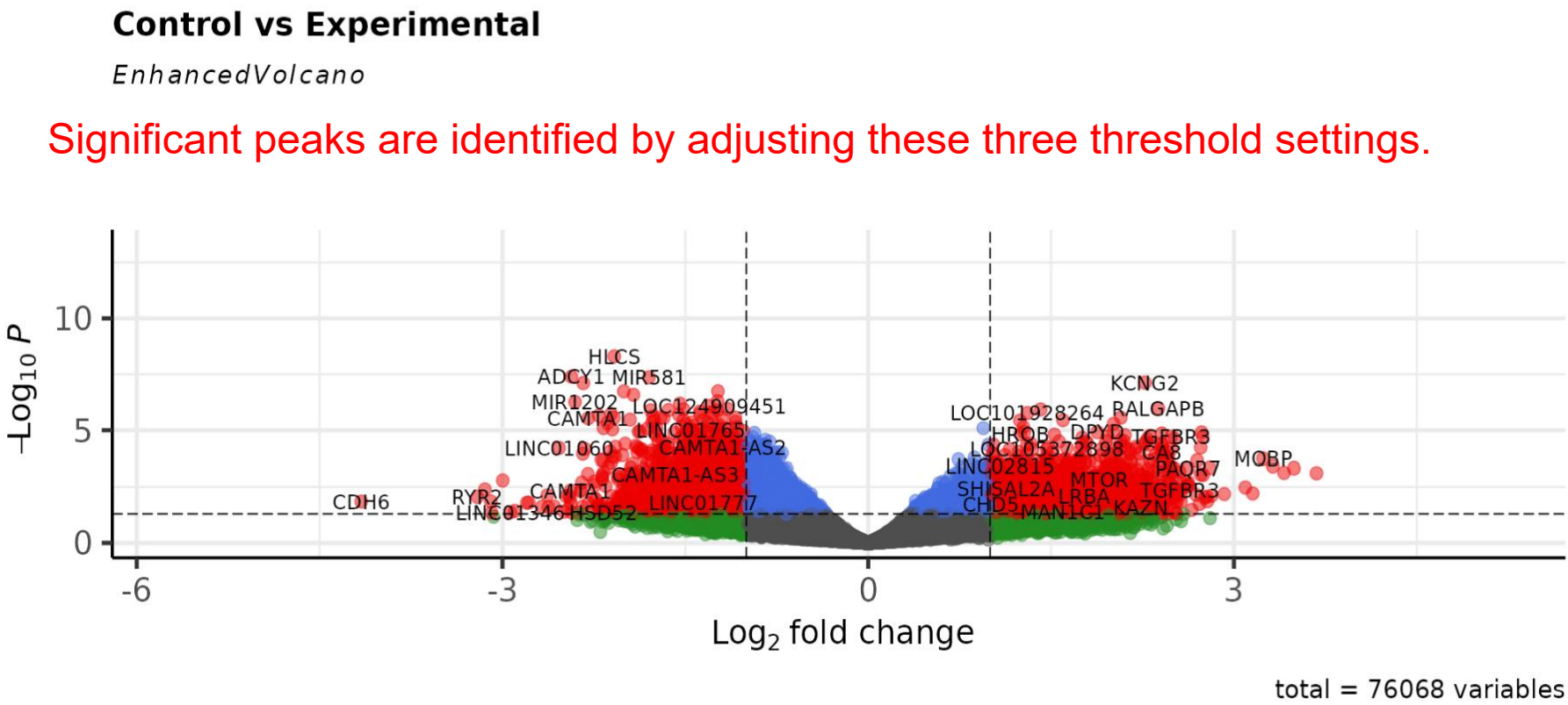
Point Size

3

Label Size

4

✓ Submit



Click the “Submit” button to update the plot and apply the selected significance criteria.

# Differential Peak Visualization Module

Up Genes Preview

Upregulated Genes List

DVL1

MRPL20

NOL9

DNAJC11

DHRS3

RPL11

ELOA-AS1

SRRM1

RSRP1

MACO1

Show 5 entries

Search:

|     | seqnames | start    | end      | width | strand | H3K27me3_rep1 | H3K27me3_rep2 | H3K4me3_ |
|-----|----------|----------|----------|-------|--------|---------------|---------------|----------|
| 17  | chr1     | 1346468  | 1350600  | 4133  | *      | 0             | 0             |          |
| 20  | chr1     | 1405327  | 1408434  | 3108  | *      | 0             | 0             |          |
| 77  | chr1     | 6552669  | 6555289  | 2621  | *      | 0             | 0             |          |
| 82  | chr1     | 6699924  | 6702352  | 2429  | *      | 0             | 0             |          |
| 142 | chr1     | 12614231 | 12620067 | 5837  | *      | 0             | 0             |          |

Showing 1 to 5 of 497 entries

Previous12345...100Next

Download

This section displays the upregulated and downregulated gene lists, as well as the peak tables generated from the filtering step.

# Functional Enrichment Module

Analysis Type:

GO

GO Ontology:

All

p-value Cutoff:

0.05

q-value Cutoff:

0.05

Upregulated Genes List

DVL1  
MRPL20  
NOL9  
DNAJC11  
DHRS3  
RPL11  
ELOA-AS1  
SRRM1  
RSRP1  
MACO1

Downregulated Genes List

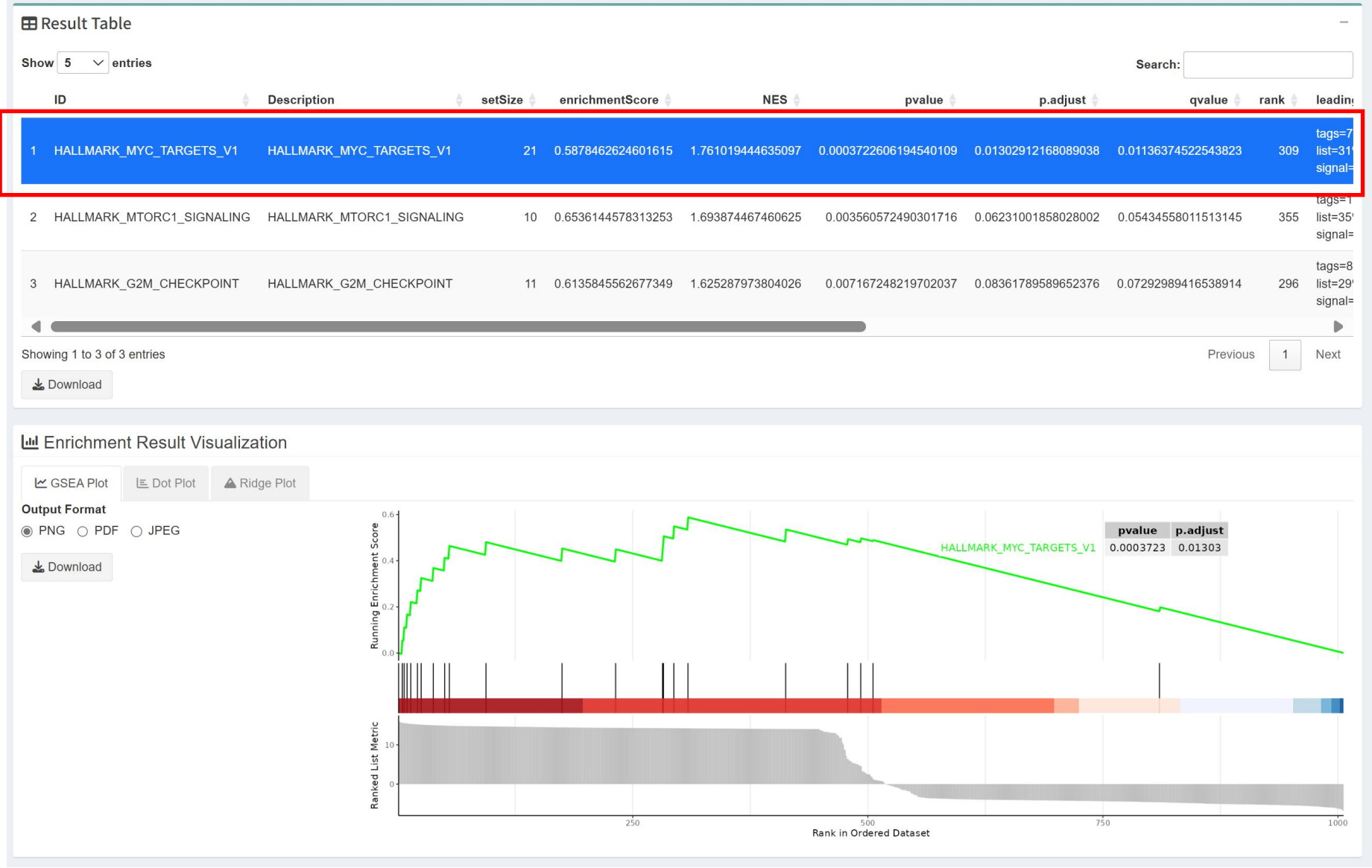
CFAP74  
MEGF6  
CAMTA1-AS3  
CAMTA1  
SLC45A1  
CA6  
SPSB1  
LINC02606  
PDPN  
KAZN

▶ Run Enrichment Analysis

Adjust the analysis type and corresponding parameters as needed.

For GO or KEGG analysis, users may enter or edit gene lists using gene symbols to define the input for enrichment testing. When GSEA is selected, the input is limited to the full set of genes annotated in the Gene Annotation module.

# Functional Enrichment Module





# Motif Enrichment Module

## Upregulated Peaks List

|                          |   |
|--------------------------|---|
| chr3-150407290-150415877 | ▲ |
| chr16-2150741-2157314    | ● |
| chr16-88972494-88978084  |   |
| chr6-119343897-119351533 |   |
| chr12-56682168-56690455  |   |
| chr8-38173859-38178882   |   |
| chr9-128688198-128692391 |   |
| chr11-65420641-65428391  |   |
| chr10-92688412-92694886  |   |
| chr6-27889944-27897392   | ▼ |
| chr11-22222222-22222222  | ✎ |

## Downregulated Peaks List

|                           |   |
|---------------------------|---|
| chr7-154746354-154754198  | ▲ |
| chr11-20155235-20165969   | ● |
| chr7-27100472-27112603    |   |
| chr1-208072367-208087339  |   |
| chr1-8323134-8331215      |   |
| chr11-118150179-118161808 |   |
| chr20-47665285-47675999   |   |
| chr14-64734713-64750256   |   |
| chr1-28810214-28820062    |   |
| chr4-182814697-182830407  | ▼ |
| chr12-118887818-11889817  | ✎ |

This section displays the top 200 upregulated and 200 downregulated peaks, which are selected from the differential peaks identified through the volcano plot filtering step. You can also input and modify your own lists of upregulated and downregulated peaks for further analysis.

## Number of Top Transcription Factors to Plot

10

## Display Motif GC Content

- ☒ Yes
- ☐ No

## Enable Clustering

- ☒ Yes
- ☐ No

- Number of Top Transcription Factors to Plot:** Set how many top transcription factors you want to visualize (default is 10).
- Display Motif GC Content:** Choose whether to show the GC content of motifs.
- Enable Clustering:** Decide whether to enable clustering for grouping similar motifs in the plot.